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AI scale and climate commitments: A 2026 outlook

Power, Energy & Electricity, Carbon Accounting, Climate Policy

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The AI and climate execution challenge

Data center energy capacity in the US is projected to increase from [25 GW to 120 GW](#) by 2030—a fivefold increase. Hyperscalers are projected to invest [\\$7 trillion](#) globally in data center infrastructure through 2030, with approximately \$2.8 trillion invested in the US.

years to develop. Evolving carbon accounting rules are shifting procurement strategies and infrastructure choices at this trillion-dollar scale, creating a “carbon debt”—embodied emissions that will stay on the books for decades. In 2026, the competitive advantage likely belongs to those who integrate power, hardware, and climate strategy from day one.

Powering AI: Grid reliability, constraints, and interconnection

Grid infrastructure faces [reliability](#) challenges from aging systems and capacity constraints. Interconnection queues stretch three to five years for renewables, while large electrical load interconnection lacks consistent standards.

Federal regulatory response

The federal government is moving to standardize these processes, with a critical decision point in 2026. For companies planning data center deployments in 2026, understanding these regulatory shifts is likely essential to realistic timeline and site selection planning.

On October 30, 2025, the US Department of Energy (DOE) leveraged Section 403(a) of the DOE Organization Act to direct the [Federal Energy Regulatory Commission \(FERC\)](#) to issue a rulemaking to “ensure efficient, timely, and non-discriminatory load interconnections” for large (>20 MW) electrical loads.

By April 30, 2026, FERC is expected to issue a final rule on [large electrical load interconnections](#) for grid operators, providing federal regulations for approval pathways, timelines, and rates.

Public comments on DOE’s advanced notice of proposed rulemaking were due on December 5, 2025, and grid operators, utilities, NGOs, and customers [submitted over 150 comments](#) reflecting a wide range of perspectives.

While federal standardization should reduce procedural uncertainty, it doesn't create new grid capacity. Even with clearer approval pathways, the underlying supply-demand mismatch remains a primary gating factor for growth.

Bridging the supply–demand gap

Data center energy demand is surging, but new clean electricity generation takes years to build. This mismatch between accelerating demand and slow-building supply is forcing the industry to pursue solutions on two timelines: near-term load flexibility strategies that

[Load flexibility](#) is emerging as a possible path to faster grid connection. Oracle, NVIDIA, Emerald AI, and Salt River Project's joint research demonstrated [25%](#) power reduction during peak hours through workload tiering. The demonstration shows that if data centers reduce consumption during peak times (roughly 1% of the year), it unlocks 126 GW of currently constrained capacity that could be available now.

Large power loads increasingly face incentives or mandates to demonstrate flexibility as part of interconnection agreements, making this an access requirement, not an optional efficiency measure. For example, Senate Bill 6 in Texas [mandates](#) that data centers and other large loads must reduce their consumption during certain grid peak times. Many other state legislatures are [passing legislation](#) that will impact data centers.

Storage has shifted from smoothing renewables to enabling multiple strategies: making intermittent renewables firmer, providing grid reliability services, and supporting 24/7 matching. Storage may emerge as a solution to allow data centers to reduce grid consumption during peak hours while maintaining operations.

[Carbon Direct helps clients](#) design load flexibility strategies under evolving regulatory frameworks: evaluating behind-the-meter generation options, sizing storage for peak reduction scenarios, and structuring interconnection configurations that preserve optionality across accounting methodologies.

Clean, firm power: Long-term generation

Hyperscalers remain committed to clean, firm generation that's reliable: power that's both low-carbon and dispatchable 24/7. [Natural gas with carbon capture and storage \(CCS\)](#) is emerging as a critical bridge technology. Google's 400 MW CCS power agreement with [Broadwing](#), expected online in 2029, demonstrates commercial demand at scale.

In our analysis evaluating [CCS pathways](#), commercial viability depends on rigorous assessment of permitting timelines, capital and operating costs, storage geology, vendor compatibility, and [45Q tax credit optimization](#). Execution has been most prevalent where technology intersects with regulatory approval and storage access.

Hyperscalers are also investing across [geothermal](#), nuclear, including Small Modular Reactors (SMR), hydrogen, and fusion. Long-duration energy storage has also been an area of focus. Each has different risk profiles and opportunities across technical maturity, permitting, commercial viability, emission accounting methodology, dispatchability, and political support.

Evaluating these pathways requires multi-dimensional frameworks. Each technology faces distinct challenges: SMRs struggle with execution complexity, geothermal with extended development periods, and hydrogen with production-dependent carbon intensity. Tax credit

Power accounting rules determine clean energy procurement

The Greenhouse Gas (GHG) Protocol extended the public consultation period for proposed [scope 2 guidance changes](#) to January 31, 2026. The results will determine clean energy procurement strategies and the carbon value of load flexibility for the next decade.

The proposed shift in electricity emissions accounting could increase clean energy procurement costs for buyers. The accounting methodological debates matter for hyperscalers: 24×7 energy matching versus carbon matching. The issues of deliverability (being located in the same grid region) and additionality (being new, rather than repurposed, generation) are also hotly debated.

These different frameworks strongly influence whether natural gas with CCS, nuclear, geothermal, or battery-backed renewables are considered optimal for a site, and whether load flexibility has carbon value.

Companies need to model scenarios across [advanced power emissions methodologies](#), evaluating portfolio costs and carbon performance before final standards are published in 2027. Companies are also signing forward renewable energy certificate contracts (RECs) and structuring power purchase agreements (PPAs) now to preserve optionality across scenarios.

AI infrastructure emissions at scale

Data center construction creates substantial scope 3 emissions, and their relative importance depends on grid carbon intensity. For facilities powered by average-carbon grids, scope 2 operational emissions dominate. But for data centers powered by very low-carbon electricity (renewables or nuclear), scope 3 embodied emissions can represent 40% of total lifetime greenhouse gas emissions.

In AI data centers, IT equipment drives the majority of embodied emissions. Chips and memory account for [67%](#), followed by structural materials at 17%, with server power supplies, aluminum, and other components comprising the final 16%.

Direct procurement of low-carbon materials faces constraints: limited supply, geographic concentration, and contracting complexity. [Environmental Attribute Credits \(EACs\)](#) provide an interim pathway by decoupling environmental benefits from physical materials, but require rigorous quality standards and verification to ensure real emissions reductions.

Carbon removal: Addressing residual emissions

Complete supply chain decarbonization by 2030 isn't feasible. Despite aggressive efforts to procure clean power and reduce construction emissions, residual emissions will remain significant. For hyperscalers with net-zero commitments, carbon dioxide removal (CDR) has shifted from an optional component to a structural necessity. Microsoft remains the world's largest CDR buyer, and Google increased purchases 14-fold from 2023 to 2024.

CDR credit quality varies widely. Companies must apply science-based principles to evaluate credits. Our [Criteria for High-Quality CDR](#), developed in collaboration with Microsoft, establishes six science-based principles for evaluating credits—critical as emerging hyperscaler and other corporate demand high-integrity supply.

AI and climate: Looking ahead

The window for strategic maneuvering is narrow. The AI infrastructure buildout is happening now, and the decisions made in 2026 will impact a company's cost structure and carbon profile for years.

Companies treating power, infrastructure, and decarbonization as separate workstreams will face compounding constraints. The winners of the AI era will be those who integrate power, infrastructure, and carbon strategy into a single, cohesive system.

REPORT

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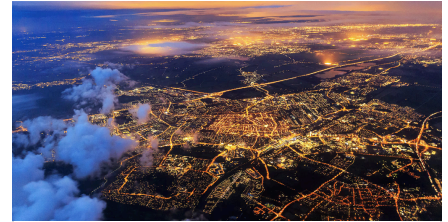
Why behind-the-meter power emissions belong in scope 2



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